RELIQ

Self-Custodial Lightning for Easy Access & Everyday Reliability

Project Foundation Document

1. Executive Summary

Reliq is a self-custodial Lightning Network platform designed for people who live and work in cash-heavy informal economies. It prioritises two non-negotiable qualities: **easy accessibility** and **everyday reliability**.

The product gives market sellers, payment agents, small traders and everyday users a simple way to send and receive value over Lightning. The interface is deliberately minimal. The system supports receiving payments even when the user's phone has intermittent connectivity. Users retain full control of their funds. A lightweight offline-capable voice layer provides balance and payment feedback in Hausa or English, making the product usable by people with limited literacy or low comfort with complex screens.

This document establishes the problem, vision, core features for the Buildathon MVP, and the chosen technical approach for offline receive. It is written to support both internal alignment and external presentation to judges, mentors or potential supporters.

2. Problem Statement

2.1 The Core Challenge

Millions of people across Nigeria and similar environments in Africa operate primarily in the informal economy. Daily commerce — market trading, small-scale services, agent-based payments and local supply chains — still depends heavily on cash and on digital tools that were never designed for their reality.

Existing digital payment systems frequently assume:

- Reliable and continuous internet connectivity
- Smartphone literacy and comfort with complex interfaces
- English-language fluency
- Trust in formal banking or custodial platforms

These assumptions leave large segments of the population underserved. The practical results are slower settlement, higher effective costs, reduced agency over funds, and continued dependence on cash even when digital alternatives theoretically exist.

The problem is not simply "lack of access to banks." It is the absence of a payment tool that is simultaneously:

- Simple enough for someone who is not tech-native
- Reliable under intermittent connectivity
- Self-custodial — the user controls their funds
- Fast and low-cost for everyday amounts
- Usable by both the person receiving money (for example a market seller) and the person sending it

Northern Nigeria provides a concrete illustration of these conditions. The underlying need, however, is not exclusive to the North. The same design principles apply wherever informal economic activity meets limited digital infrastructure and low tolerance for complex tools.

2.2 Why Current Solutions Fall Short

Most existing Lightning or Bitcoin wallets prioritise power users or assume consistent connectivity and higher technical comfort. Custodial fintech applications improve convenience for some users but introduce counterparty risk and often remain English-first, with interfaces that still feel foreign to many market participants. There is a clear gap for a self-custodial Lightning experience that treats simplicity, reliability under real-world conditions, and local-language accessibility as first-class requirements.

3. Product Vision

Reliq is a self-custodial Lightning Network platform designed for easy accessibility and everyday reliability. It enables individuals and small agents in cash-heavy informal economies to send and receive value quickly, with an interface simple enough for non-tech-native users, support for intermittent connectivity, and a lightweight offline-capable voice layer that speaks key information in Hausa or English.

3.1 Guiding Philosophy

- Ease of access first — The interface must feel obvious. Large, clear actions. Minimal cognitive load.
- Reliability under real conditions — Designed for intermittent connectivity rather than assuming perfect internet.
- Self-custody by default — Users control their funds. No third party holds the keys.
- Lightning as the settlement layer — Fast, low-cost movement of value.
- Local accessibility — Voice feedback and language support so the product works for people with limited literacy or low comfort with complex screens.

3.2 Primary Users

1. Market sellers and small traders who need a simple way to receive payments and keep a clear record.
2. Payment agents / POS-style operators who need reliable inbound liquidity and settlement visibility.
3. Everyday users who want to send or receive small amounts without depending on complex banking flows.
4. Users with limited literacy or preference for voice interaction who still need reliable digital payments.

4. Core Features — MVP Scope

The Buildathon version focuses on a working, demonstrable system. Scope is deliberately tight so the product remains simple, reliable and easy to demo.

4.1 Primary Experience — Simplified Mobile Application

The user-facing product is a mobile application with an extremely clear interface:

- Large, obvious actions: Receive, Send, Balance, Notifications / Activity
- Minimal text and strong visual hierarchy
- Receiver shares a simple Lightning Address, QR code, or short unique identifier

- Buyer pays by scanning or entering the identifier
- Seller receives a clear notification and updated balance

4.2 Reliability Under Intermittent Connectivity

Reliq uses a production-proven approach based on the Breez SDK model (webhook + push notification) for offline and intermittent receive:

- The user's phone does not need to remain constantly online to receive payments
- When a payment arrives, a webhook triggers a push notification that wakes the app
- The app claims the payment and updates the balance
- Keys remain on the user's device — the solution stays self-custodial

This approach is already used in production by multiple applications and is significantly more mature than pure LDK Async Payments for a Buildathon timeline and real-world reliability.

4.3 Offline-Capable Voice Layer (Accessibility & Localization)

A lightweight, offline-capable voice layer improves accessibility and local relevance:

- User can hear balance, payment received, and basic status messages
- Supports Hausa and English
- Designed to work with limited or intermittent connectivity (on-device or lightweight offline TTS where practical)
- Triggered by a clear button or simple interaction so it does not complicate the core UI

4.4 Explicitly Out of Scope for MVP

The following are deliberately excluded from the current Buildathon scope to keep the product focused and demoable:

- Bluetooth / NFC / local wireless mesh for fully offline peer-to-peer payments
- Full production USSD as the primary interface
- Complex multi-agent or cross-border settlement systems beyond basic demonstration flows

5. Positioning Summary

Reliq solves for easy accessibility and reliability in environments where cash still dominates and existing digital tools remain too complex or too dependent on perfect connectivity.

It does this by combining:

- A radically simplified mobile interface
- Self-custodial Lightning with production-proven intermittent-connectivity support (Breez-style webhook + push)
- A lightweight offline-capable voice layer in Hausa and English
- A deliberately tight MVP scope that prioritises clarity and reliability over feature breadth

The result is a product that is immediately useful, demonstrable within the Buildathon timeline, and strategically positioned for deeper technical contribution after the event.